Variable Admittance Control Using Velocity-Curvature Patterns to Enhance Physical Human-Robot Interaction

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***Abstract*—This letter introduces a variable admittance control approach aimed at enhancing intuitive human-robot interaction by considering both direct and indirect human intentions. The magnitude of force serves as a representation of direct intentions, delineating preferences for rapid or precise motions. Drawing from the movement sciences and motor control field, the minimum-jerk model is employed to mirror human motor system control policies and movement behaviors. From this model, velocity-curvature patterns are derived, enabling an intuitive estimation of indirect intentions indicating long-term objectives like trajectory and turning direction. We propose an innovative guidance method, rooted in the estimation of indirect human intentions, enabling the robot to concurrently follow and guide the operator. To assess the efficacy of this approach, both offline simulations and real-time human experiments are conducted on a six-DOF robot and a force/torque sensor. Our comprehensive experiments demonstrate substantial enhancements in both accuracy ($\geq 12.1\%$) and smoothness ($\geq 18.3\%$) over fixed admittance control and state-of-the-art variable admittance methods.**

***Index Terms*—Human factors and human-in-the-loop, intention recognition, physical human-robot interaction, velocity-curvature patterns.**

I. Introduction

PHYSICAL human-robot interaction (pHRI) becomes increasingly important as the futuristic vision of robots working with humans tends to be a reality. Specifically, pHRI is a promising application in various fields such as assembly, teleoperation, robotic rehabilitation, and others. In recent years, the popularity of using lightweight robots and collaborative robots for pHRI has grown due to their affordability, improved working environments, and efficiency.

Direct teaching is a well-known function of pHRI where people drag the end of the robot to generate motion trajectories. Specifically, direct teaching has promising applications in transportation [1], welding [2], [3], polishing [4] and others. Most functional tasks mentioned above could be simplified as a trajectory tracking task. Admittance control is commonly used in direct teaching for its active compliance [5]. In direct teaching, the robot needs to be controlled as an admittance model in most scenes because the force is generated by humans and the robot needs to output corresponding displacement to comply with the force input [6].

Determining human intent is an essential part of direct teaching to enhance intuitiveness. In a previous study [7], human intention is divided into direct intention and indirect intention. Direct intention indicates the instant response from the human force. In [8], Ficuciello et al. vary the damping according to the absolute value of end-effector Cartesian velocity to improve the performance in terms of execution time and accuracy. Lecours et al. propose a variable admittance control law adjusting the admittance parameters based on the magnitude of acceleration in [9]. Similar schemes [10], [11] is proposed to update the admittance law parameters according to the input force. When faster motion is required which means larger force input, the robot becomes more compliant. However, in [10] the reference path needs to be known in advance to implement the algorithm of virtual stiffness guidance. In [12], the product of Cartesian velocity and acceleration, representing the change in kinetic energy, was utilized to transition between negative and positive damping. However, in [12], only simple point-to-point target reaching task is considered. Additionally, when faced with more complex trajectory tracking tasks, negative damping may cause instability.

Indirect intention implies a long-term purpose of human such as straight or curve motion. It is proved that considering the indirect intention will further improve the smoothness than considering the direct intention alone [7]. Because although the estimation of direct intention provides intuitiveness, it also

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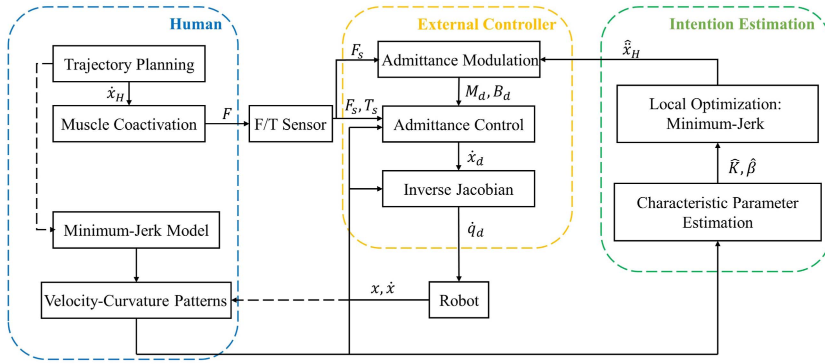

Fig. 1. Block diagram of the proposed variable admittance control with human intention estimation during direct teaching. Decode the magnitude of the force F_s as the direct intention of the human: a desire for rapid or precise motion; while considering the previous path as indicative of the indirect intention of humans $\hat{\dot{x}}_H$: trajectory and turning direction. Our proposed variable admittance control achieves intuitive and real-time guidance functionality by simultaneously considering these two intentions. $\hat{K}$ and $\hat{\beta}$ represent the estimated characteristic parameters of the path.

includes sensor noise and human errors induced by noise. Not only the motor command but also the generated force is affected by noise [13]. Due to the noises, humans are unable to move their hands along a precise expected trajectory with unexpected errors occurring at times. To compensate for the error and noise, researchers are tempted to develop a variable admittance control law to assist humans by guiding the robot along an expected path in real-time. Virtual fixtures [14], [15], [16] are used to solve the similar problem in haptic systems according to long-term intention. The virtual fixtures are applied to guide the human moving along a predefined path or staying away from certain regions. However, studies using this method always use predefined paths or regions which ignore human online indirect intention. To combine the real-time human indirect intention into the controller, the product of Cartesian velocity and interaction force is used to represent the direct intention of humans, and the cosine value of the angle between the interaction force and the coordinate axis is used to represent indirect intention [17]. A fuzzy rule integrating two intents shows better performance in terms of execution time, integrated squared jerk, and path error than the rule only with the direct intention. However, the approach in [17] fails to provide a continuous estimation of the direction of indirect intention and the estimated values are unable to cover the entire directional space. In [18], a neural network is proposed based human motion intention estimator to predict the trajectory of hand movement, the motion intention is regarded as a function that is influenced by human force, velocity, and position. In [7], it is assumed that indirect human intention corresponds to a curve on a plane with a constant curvature that includes a straight line. The reference variable is proposed to form the expected force to guide force direction. As a result, the force guidance controller significantly improves the smoothness of the motion compared with the controller only considering the direct intention. However, contemporary methods require training, setting empirical parameters in advance, or relying on simple assumptions without a physiological basis. These requirements make the preparation of direct teaching longer and more cumbersome, and to some extent do not reflect the true indirect human intentions.

In this study, a universal and training-free variable admittance controller is introduced for a position-controlled manipulator, enabling free motion and precise plane motion based on the human intention estimation as it is shown in Fig. 1. The controller incorporates the minimum-jerk model and velocity-curvature patterns, well-known as the control policy and resulting movement phenomenon, respectively. The constraint on the intended acceleration is achieved by estimating the characteristic parameters of the path based on velocity-curvature patterns. The adapted minimum-jerk is employed to determine both the magnitude and the orientation of the intended acceleration.The controller outputs the intended force as the intended direction. Average trajectory error, dimensionless squared jerk and mean force are quantified to demonstrate the overall performance of this approach compared with the fixed admittance control, the isotropic variable admittance method in [11] (referred to as the isotropic method), the fuzzy variable admittance control (FVAC) in [17] and the variable admittance method in [7] (referred to as Const. Rc).

In summary, our primary contributions are: (1) A novel indirect human intention estimation based on movement sciences and motor control results such as minimum-jerk model and velocity-curvature patterns. (2) A universal and training-free variable admittance controller that outputs intended force to provide real-time haptic guidance in direct teaching without requiring a predefined path.

II. Method

A. Background

1) Admittance control model: The dynamic model of admittance control in Cartesian space is a mass-spring-damper system, which can be expressed as:

$$\boldsymbol{M}_d\left(\ddot{\boldsymbol{X}}-\ddot{\boldsymbol{X}}_0\right)+\boldsymbol{B}_d\left(\dot{\boldsymbol{X}}-\dot{\boldsymbol{X}}_0\right)+\boldsymbol{K}_d\left(\boldsymbol{X}-\boldsymbol{X}_0\right)=\boldsymbol{F}_{ext} \tag{1}$$

where $\boldsymbol{M}_d$, $\boldsymbol{B}_d$, $\boldsymbol{K}_d$ represent virtual mass, damping and stiffness matrices. Additionally, $\boldsymbol{X}$, $\dot{\boldsymbol{X}}$, $\ddot{\boldsymbol{X}}$ represent the position, velocity, and acceleration. As this letter focuses on the planar movement, $\boldsymbol{X}$ denotes a two-dimensional vector. $\boldsymbol{X}_0$ indicates

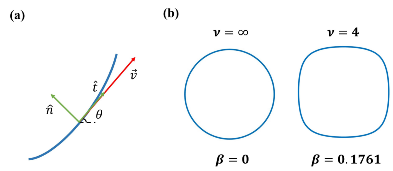

Fig. 2. (a) Frenet-Serret frame is used and the angular direction of the tangent vector θ is regarded as the new coordinate along the path. (b) Examples of pure frequency curves with their characteristic frequencies ν. For example, a squircle path is characterized by $\nu = 4$, because its curvature profile oscillates four times per one rotation of θ. In contrast, a circular path retains a consistent curvature, thus its characteristic frequency ν is considered infinite.

the equilibrium point of the robot. $\boldsymbol{F}_{ext}$ indicates the external force of the environment.

In direct teaching, human transmits information concerning the path to the robot, information that remains undisclosed to the robot. So the virtual spring matrix $\boldsymbol{K}_d$ and equilibrium point $\boldsymbol{X}_0$ are set to zero. Therefore the dynamic model of admittance control is reshaped as a mass-damper system:

$$\boldsymbol{M}_d \ddot{\boldsymbol{X}} + \boldsymbol{B}_d \dot{\boldsymbol{X}} = \boldsymbol{F}_{ext} \tag{2}$$

2) Minimum-jerk model and spectrum of power laws for curved hand movements: In [19], Flash and Hogan formulate a mathematical model which predicts both the qualitative features and the quantitative details observed experimentally in planar, multi-joint arm movements:

$$\min \int_0^{t_f} \left(\dddot{x}(t)^2 + \dddot{y}(t)^2 \right) dt \tag{3}$$

In natural drawing movements, numerous studies have found a systematic relationship between the kinematics of the hand motion and the geometric properties of the path [20], [21], [22]. This principle of human movement, referred to as the two-thirds power law, is described as:

$$v(t) = K\kappa(t)^{-\beta} \tag{4}$$

where v is the tangential hand velocity, κ is the curvature of the trajectory, K is the velocity gain factor which depends on the length of the segment, $\beta = 1/3$. This power law describes the observation that the velocity of hand movement decreases when the trajectory becomes more curved and increases when the trajectory becomes straighter.

Moreover, Viviani and Flash prove that the kinematics and geometrical parameters of the planar movement formalized by two-thirds power law can be derived from the minimum-jerk model hypothesis [23]. Furthermore, Huh investigates the speed and curvature phenomenon for more general curved hand movements by using the minimum-jerk model [24]. Huh introduces the orientation of the tangent vector as the new coordinate system, the angle coordinate θ (Fig. 2(a)). Pure frequency curves that have sinusoidal log curvature profiles are defined (Fig. 2(b)):

$$h(\theta) = \epsilon \sin(\nu\theta) \tag{5}$$

where $h(\theta) = log(\kappa/\kappa_0)$ is the log curvature profile with arbitrary constant κ_0, ν is the frequency and ϵ is the amplitude. The analysis confirms the power law with exponent $\beta = 1/3$ for drawing ellipses but also a spectrum of power laws with exponents β ranging between 0 and $2/3$ for simple movements that can be characterized by a single angular frequency ν. In this study, the mixture of power laws is also confirmed by human drawing experiments for more complex curves.

$$\beta(\nu) = \frac{2}{3}\left(\frac{1 + \nu^2/2}{1 + \nu^2 + \nu^4/15}\right) \tag{6}$$

In [25], it has been proved that considering the human velocity-curvature patterns facilitate pHRI in the circumstance of elliptical movement where $\beta = \frac{1}{3}$. It is inspired that velocity-curvature patterns may have the potential to be extended to more complex trajectories in pHRI to achieve intuitive and efficient interaction.

B. Proposed Variable Admittance Control

There are three steps for the proposed variable admittance control method. In the first step, the magnitude of force is regarded as direct intention to tune the initial damping parameter in an isotropic way. In the second step, least-square method is used to estimate the parameters of the power law which reveals the characteristic of the preceding long-term path. The intended acceleration is computed by combining the estimated power law parameters with the adapted minimum-jerk, establishing an optimal objective. Finally, deriving the intended force from the intended acceleration, the guidance method is implemented along the direction indicated by the intended force.

1) Direct intention: Adapting variable admittance for direct intention involves updating damping parameters based on the magnitude of force, offering greater intuitiveness compared to velocity. This approach ensures users' enhanced control and comfort. When precision is needed, users apply minimal force, resulting in a somewhat rigid robot response. Conversely, for swift motions, larger forces induce a compliant robot behavior. Determining the force range $[f_{\min}, f_{\max}]$ involves experimental assessments by multiple users performing specific tasks. Consequently, the initial damping parameter b_d correlates linearly with the magnitude of force ($f = ||F_{ext}||$).

$$b_d = \begin{cases} b_{\max} & f \leq f_{\min} \\ b_{\max} - \frac{b_{\max} - b_{\min}}{f_{\max} - f_{\min}}(f_{\max} - f) & f_{\min} < f < f_{\max} \\ b_{\min} & f \geq f_{\max} \end{cases} \tag{7}$$

2) Indirect intention estimation: Prior to estimating indirect intention, it is well-observed that when a human tempts to draw a straight line, the presented drawing cannot be perfectly straight. Thus it is assumed that intended linear movement, in reality, can be regarded as curved movement to some extent.

In real-time direct teaching, it is unrealistic to obtain the whole path information in advance. Using the fast Fourier transform to derive ν from the curvature of the previous path also fails to meet the real-time operational requirements.. Therefore, least-square method is introduced to obtain the parameter β and K of the specific power law corresponding to the present path.

The velocity vector $\hat{\boldsymbol{v}}(k) = [\hat{\boldsymbol{v}}_x(k), \hat{\boldsymbol{v}}_y(k)]^T$ for the present period is calculated with forward finite difference approximations. The velocity vector $\hat{\boldsymbol{v}}(k-n)$ and acceleration vector

$\hat{\boldsymbol{v}}(k-n)$ for the previous n periods are obtained with central finite difference approximations. The curvature of the planar curve of the previous n periods is calculated as:

$$\hat{\kappa}(k-n) = \frac{\left|\hat{\boldsymbol{v}}_x(k-n)\dot{\hat{\boldsymbol{v}}}_y(k-n) - \dot{\hat{\boldsymbol{v}}}_x(k-n)\hat{\boldsymbol{v}}_y(k-n)\right|}{\|\hat{\boldsymbol{v}}(k-n)\|_2^3} \tag{8}$$

With the curvature and velocity within the past N periods obtained, the parameters of the power law can be estimated using least-square method by taking logarithms on both sides of (4) as:

$$\begin{bmatrix} 1 & ln\hat{\kappa}(k-N) \\ \dots & \dots \\ 1 & ln\hat{\kappa}(k-1) \end{bmatrix} \begin{bmatrix} ln\hat{K} \\ -\hat{\beta} \end{bmatrix} = \begin{bmatrix} ln\|\hat{\boldsymbol{v}}(k-N)\|_2 \\ \dots \\ ln\|\hat{\boldsymbol{v}}(k-1)\|_2 \end{bmatrix} \tag{9}$$

By substituting the estimated parameter of $\hat{K}$ and $\hat{\beta}$ along with the present velocity $\hat{\boldsymbol{v}}(k)$ into (4), we derive an equation that represents the intended acceleration $\dot{\boldsymbol{v}}_d(k)$. This equation serves as the primary constraint in (10).

As discussed in the previous section, the minimum-jerk model is regarded as the path-planning policy in the human system. Since (4) only reveals the correlation between the magnitude of velocity and acceleration which lacks the information of orientation, so minimum-jerk is adopted to acquire the orientation and magnitude of the intended acceleration. A local optimization method adapted from the minimum-jerk model is proposed to solve the $\dot{\boldsymbol{v}}_d$ as follows:

$$\begin{aligned} &\underset{\dot{\boldsymbol{v}}_d(k)}{\text{minimize}} \left\| \frac{\dot{\boldsymbol{v}}_d(k) - \dot{\hat{\boldsymbol{v}}}(k-1)}{T} \right\|_2^2 T \\ &\text{subject to } |\hat{\boldsymbol{v}}(k) \times \dot{\boldsymbol{v}}_d(k)|^{-\hat{\beta}} = \frac{\|\hat{\boldsymbol{v}}(k)\|_2^{1-3\hat{\beta}}}{\hat{K}}, \\ &\qquad (\hat{\boldsymbol{v}}(k) \times \boldsymbol{F}_{ext}(k))(\hat{\boldsymbol{v}}(k) \times \hat{\boldsymbol{v}}(k+1)) > 0, \\ &\qquad \frac{\hat{\boldsymbol{v}}(k)\hat{\boldsymbol{v}}(k+1)}{\|\hat{\boldsymbol{v}}(k)\|\|\hat{\boldsymbol{v}}(k+1)\|} > \frac{\hat{\boldsymbol{v}}(k)\boldsymbol{F}_{ext}(k)}{\|\hat{\boldsymbol{v}}(k)\|\|\boldsymbol{F}_{ext}(k)\|}, \end{aligned} \tag{10}$$

where $\hat{\boldsymbol{v}}(k+1)$ denotes the intended velocity of the next period which is derived from forward Euler method as $\hat{\boldsymbol{v}}(k) + \dot{\boldsymbol{v}}_d(k)T$. The optimization objective is derived from (3) based on the assumption that during the small period $[(k-1)T, kT]$, the jerk $\ddot{\boldsymbol{v}}(t)$ can be approximated with forward finite difference method as $(\dot{\boldsymbol{v}}_d(k) - \dot{\hat{\boldsymbol{v}}}(k-1))/T$. The last two constraints are simplified versions of the dynamic constraints, representing that the direction of the velocity at the next period should be in the angle between the direction of the force and the velocity at the current period.

With the intended acceleration obtained, the intended force for the next iteration can be derived from (2) as follows:

$$\hat{\boldsymbol{F}}_{intended} = m_d\dot{\boldsymbol{v}}_d(k) + \frac{b_d}{\gamma_1}\hat{\boldsymbol{v}}(k) \tag{11}$$

where m_d is the default virtual mass value and γ_1 is introduced to encourage the motion in the intended direction which will be discussed in the next section. The intended force consists of two components: the first component represents the force required

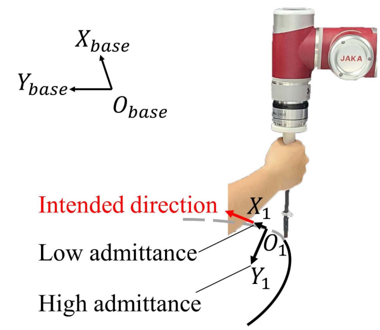

Fig. 3. Scheme of the variable admittance parameters based on intended direction from intended force. The direction of the intended force is regarded as the intended direction X_1, and the reduction of admittance parameters along this intended direction enables better online guidance.

to change direction, calculated from the intended acceleration obtained from (10), and the second component represents the minimum driving force. The intended force is derived from long-term intention estimation which makes the interaction more intuitive and smoother than those obtained directly from the sensors which may include much more noise from devices and environment. It is studied that there is a limitation in human motions in terms of a frequency band, which can be assumed to be from 0.5 to 5 Hz [26], [27]. In [28], Kalinowska chooses 2.5 Hz as the highest frequency in human-robot interaction. Due to the frequency limit of the human-robot interaction, both the $\hat{\boldsymbol{F}}_{intended}$ and the output of Const. Rc method are filtered with a low-pass Butterworth filter and a cutoff frequency of 5 Hz.

3) Variable admittance control with guidance: Fig. 3 shows the basic concept of the guidance method. The intended force have their certain direction when the operator interacts with the robot to move along an intended path. If the robot can be moved in the intended direction easily while is hard to be moved along the direction that is vertical to the intended one, the robot controller will help the operator reduce error, increase interaction comfort and decrease effort.

The intended force are obtained from the indirect intention estimation which is explained earlier. To guide the direction of the intended force, lower admittance parameters are set to the direction of intended force which is regarded as the intended direction, and higher admittance parameters are set to the direction which is vertical to the intended one. If x-axis corresponds to the intended direction, the virtual damping matrix is changed as follows:

$$\boldsymbol{B} = \begin{bmatrix} \frac{b_d}{\gamma_1} & 0 \\ 0 & \gamma_2 b_d \end{bmatrix} \tag{12}$$

where $\boldsymbol{B} \in \mathbb{R}^{2\times 2}$ denotes the virtual damping matrix in the planar Cartesian space. It is pertinent to highlight that the selection of the intended direction can readily encompass three dimensions. For instance, the restructuring of the damping matrix (12) like $diag\left(\frac{b_d}{\gamma_1}, \gamma_2 b_d, \gamma_2 b_d\right)$ serves as an exemplification.

Lower admittance parameters which are set in x-axis decrease as speed increases. Higher admittance parameters increasing

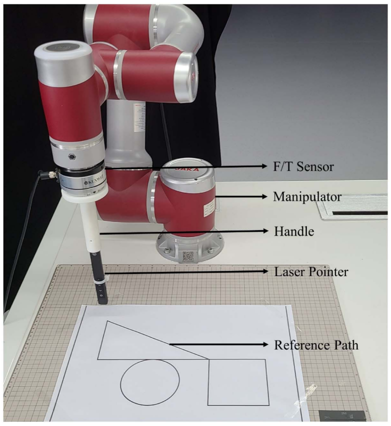

Fig. 4. Experiment setup. Operators drag the handle along the reference path in a horizontal plane. The laser pointer is used to provide visual feedback.

proportionally to the norm of the velocity are set in y-axis:

$$\gamma_1 = 1 + 2\|\hat{\boldsymbol{v}}(k)\|_2, \gamma_2 = 1 + 4\|\hat{\boldsymbol{v}}(k)\|_2 \tag{13}$$

The x-axis changes with respect to the intended direction and $\boldsymbol{R} \in \mathbb{R}^{2\times 2}$ denotes the rotation matrix between the robot base frame and the x-axis frame based on the intended direction. Finally, the virtual damping matrix in the robot base frame is calculated as follows:

$$\boldsymbol{B}_d = \boldsymbol{R}\boldsymbol{B}\boldsymbol{R}^T \tag{14}$$

III. Experiments

A. Experimental Setup

As shown in Fig. 4, the experimental system consists of a six-DOF collaborative robot (JAKA ZU7) and a force/torque sensor (KWR75B KUNWEI) attached to the end effector of the robot. During experiments, a laser pointer displays the handle position in real-time on the horizontal plane. The robot's joint angles and force/torque sensor signals are recorded by the controller at a sampling rate of 125 Hz and transmitted via Ethernet.

Fig. 1 shows the control scheme of the variable admittance control with intention estimation. Operators maneuver the robot by applying force to the sensor on the end effector. In both experiments, the end effector's orientation remains vertical to the horizontal plane, with z-axis position unchanged. Cartesian velocity, obtained via variable admittance control, is converted to joint angular velocities using the inverse Jacobian matrix. This information is then relayed as movement commands to the robot controller. Here the velocity $\dot{x}_d$ is regarded approximately equal to the actual robot velocity $\dot{x}$ for the well-tuned internal controller. The repeatability of the position controller is ± 0.02 mm. According to experimental verification, the controller can reach the desired position within 4 cycles under the action of the simulated step force signal.

B. Experiment Task

The reference path in Fig. 4 includes straight lines of 0.1 m length connecting the circular and square paths, with a 0.225 m distance between the square path and the starting position. The circular path's diameter and the square path's side length are both set to 0.1 m, as adapted from [7].

In the offline simulation, an operator drags the end effector of the robot along the reference path in the fixed admittance method where the $\boldsymbol{M}_d$ and $\boldsymbol{B}_d$ are set to be diagonal matrices as $diag(m_d, m_d)$ and $diag(b_d, b_d)$. The default virtual mass and damping correspond to $m_d = 5$ kg and $b_d = 120$ Ns/m, respectively. The number of cycles N used in least-square method is selected as 50. The intended direction of force is simulated using the recorded data. The intended direction is compared with the force direction which is assumed to be the operator's real intended direction to evaluate the efficiency and accuracy of the proposed indirect intention estimation method.

During the online experiment, operators aim for accurate end effector movement within 25 seconds using five controllers. The range of damping values across the main axes in all variable admittance controllers is [90,150] Ns/m, while the damping of the fixed admittance controller remains 120 Ns/m. The isotropic method [11] is capped at a maximum force of 11 N, and the absolute value range of the power of interaction for the FVAC method [17] is set from 0 to 2 W. Another method is referred to as Const. Rc in [7]. Both the force guidance method based on indirect intention and direct intention method is composed in Const. Rc where the standard force in direct intention is set as 8 N. In our proposed variable admittance controller, the maximum $f_{\max}$ and minimum $f_{\min}$ values of force are set as 11 N and 1 N. Every operator practices for 5 min under fixed admittance control. After the practice, the operator gets familiar with each method for 3 trials. After the familiarization and practice, the operator repeats the experiment eight times with each method. In the online experiment section, the order of different methods is randomly selected, keeping both the operator and experimenter unaware of the ongoing experiment's method sequence.

Fifteen healthy subjects (age: 22–29, height: 162–183 cm, weight: 48–100 kg, sex: 11 males and 4 female) participate in this study, which is approved by the Institutional Review Board of Shanghai Jiao Tong University (Approval No. : E20230258I). Subjects provide informed, written consent prior to participation. All experimental procedures are performed in accordance with the relevant guidelines and regulations. No subject is informed of the hypotheses of this study.

C. Data Analysis

Several performance metrics representing accuracy, smoothness, effort are selected to compare the proposed controller with a fixed admittance controller, isotropic method, FVAC and Const. Rc method. Forward finite difference approximation is applied to calculate the speed at each time step in every trial data. A threshold of 0.005 m/s is set to differentiate the movement data from the whole trial data.

1) Accuracy: Accuracy is evaluated by the average error between the robot movement path in direct teaching and the reference path in the horizontal plane. The average error in one trial which contains M time steps movement data is defined as:

$$Error = \frac{1}{M}\sum_{i=1}^{M} \|\boldsymbol{X}(i) - \boldsymbol{X}_{reference}\|_2 \tag{15}$$

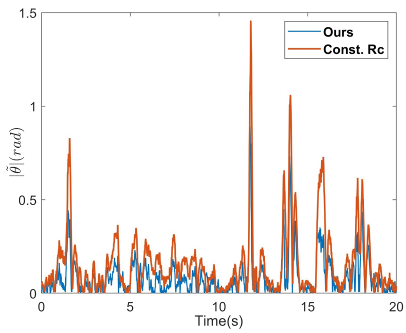

Fig. 5. Offline simulation results. The real-time force orientation is regarded as the indirect human intention. The evaluation metric is the absolute error of the direction.

TABLE I
REGRESSION RESULT OF THE OFFLINE SIMULATION

	Const. Rc	Ours
intercept	0.0473	**0.021**
RMSE	0.2271	**0.1265**

2) Smoothness: Smoothness is evaluated by dimensionless squared jerk (DSJ) [29]. Jerk, the time derivative of acceleration, has been used as an empirical measure of the smoothness of movement. The DSJ is defined as:

$$DSJ = \int_{t_1}^{t_2} \left(\dddot{x}(t)^2 + \dddot{y}(t)^2\right) dt \frac{(t_2 - t_1)^5}{A^2} \tag{16}$$

where parameter A is the total length of path $X(i)$ taken by the end effector of robot and human hand from start time t_1 to final time t_2.

3) Effort: Effort is evaluated by the average 2-norm of human-robot interaction force in direct teaching which will be referred to as mean force. The mean force is defined as:

$$MeanForce = \frac{1}{M} \sum_{i=1}^{M} \|\boldsymbol{F_{ext}}(i)\|_2 \tag{17}$$

IV. Results

To evaluate the effectiveness of the proposed variable admittance control with indirect intention estimation, an offline simulation is conducted to compare the proposed method with Const. Rc method. In the online experiment, five different methods are compared in real-time with their performance metrics. Graphic results are obtained and statistical results are compared using t-test.

A. Results of the Offline Simulation

As can be seen in Fig. 5, the proposed indirect intention estimation which outputs the intended force exhibits significantly smaller prediction error. A one-dimensional linear regression relates the direction of recorded force to the output direction from indirect intention estimation using the Curve Fitting Toolbox of MATLAB in Table I. The fitted slopes of different methods are all adjusted to 1 to be convenient for comparison.

From the offline simulation, the proposed intention estimation method predicts better than Const. Rc method.

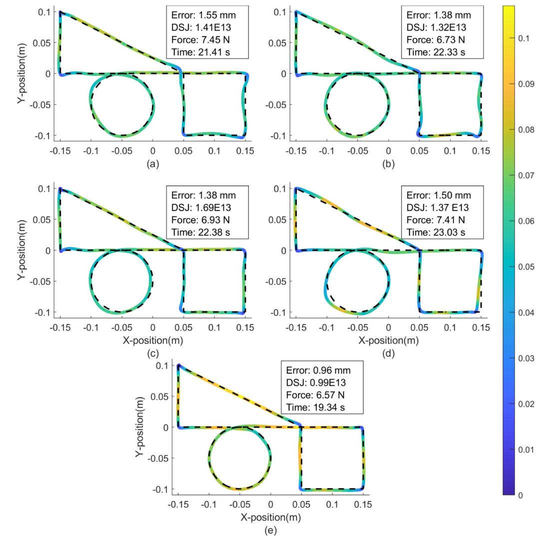

Fig. 6. Examples of a typical subject 11 in the online experiment: (a) fixed admittance control; (b) isotropic admittance control; (c) FVAC; (d) const. Rc method; (e) ours controller.

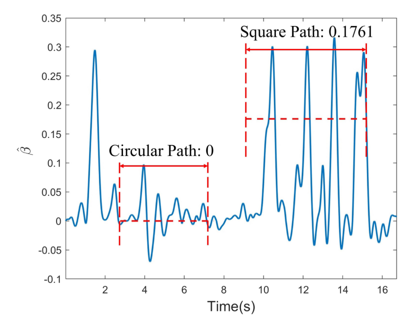

Fig. 7. Example of the estimated parameter β (low-pass filtered) in our proposed controller. The red bidirectional arrows show the duration of the circular path and the square path. The 0 and 0.1761 is the ideal parameter calculated from (6) based on the characteristic frequency illustrated in Fig. 2.

B. Results of the Online Experiment

In Fig. 6, examples with a single subject are illustrated with each admittance controller. The human-robot cooperation is illustrated by the responding error, jerk and mean force.

In Fig. 7, the estimated parameter $\hat{\beta}$ in Fig. 6(e) is shown. For a circle, the curvature remains constant at all times and the frequency ν can be considered to be infinite. For the square path, it is similar to the curve ($\nu = 4$) which is illustrated in Fig. 2(b). As can be seen in Fig. 7, the estimated parameter $\hat{\beta}$ fluctuating around the ideal value means that the prediction of the shape in our proposed method is accurate, thus explaining the advantage of our proposed controller in Fig. 6. Similar to the offline simulation, the proposed method exhibits significantly lower average absolute estimation errors in online experiments compared to Const. Rc (0.137 vs. 0.242 rd, $p < 0.001$).

The proposed variable admittance controller demonstrates superior performance across all metrics, encompassing the entirety of the path as well as its individual circular and square segments, as delineated in Table II. Fig. 8 visually encapsulates the performance metrics across various strategies. Notably, the proposed variable admittance controller showcases

TABLE II
COMPARISON OF PERFORMANCE FOR CONTROLLERS

	Fixed	Isotropy	FVAC	Const. Rc	Ours
Error (mm)					
- whole path	1.43	1.35	1.28	1.45	**1.11**
- circular path	1.51	1.41	1.37	1.70	**1.13**
- square path	1.58	1.49	1.39	1.55	**1.22**
DSJ					
- whole path(E13)	1.28	1.19	1.18	1.27	**0.84**
- circular path(E10)	6.06	5.54	5.81	6.20	**4.11**
- square path(E11)	1.54	1.59	1.43	1.63	**1.16**
Mean force (N)					
- whole path	7.31	6.85	7.10	7.83	**6.64**
- circular path	7.09	6.86	6.99	7.91	**6.60**
- square path	7.20	6.54	6.77	7.36	**6.15**
Task time (s)					
- whole path	21.93	21.64	21.66	21.93	**20.00**
- circular path	5.80	5.71	5.79	5.83	**5.37**
- square path	7.14	7.22	7.07	7.29	**6.81**

substantial enhancements–13.5% ($p < 0.01$), 29.2% ($p < 0.001$) and 3.1% ($p < 0.05$) –in accuracy, smoothness, and effort reduction, respectively, compared to the most proficient among the comparative methods, encompassing the entire path. Moreover, within the circular path, this controller yields gains of at least 17.6% ($p < 0.001$), 25.7% ($p < 0.001$) and 3.9% ($p = 0.06$) in accuracy, smoothness, and effort reduction when contrasted with alternative methods. Specifically focusing on the square path, the proposed controller consistently enhances accuracy, smoothness, and effort reduction by 12.1% ($p < 0.05$), 18.3% ($p < 0.05$) and 5.9% ($p < 0.001$), respectively, when juxtaposed with other methods.

V. DISCUSSIONS

This letter introduces a variable admittance controller incorporating both direct and indirect estimation of human intention, particularly relevant in direct teaching scenarios. Our proposed method demonstrates substantial enhancements in tracking accuracy, smoothness and effort reduction compared with the fixed admittance controller and three state-of-the-art variable admittance controllers. The isotropic approach's limitations lie in focusing solely on the human's direct intention, disregarding human directional preferences during trajectory tracking. The FVAC method fails to provide a continuous estimation of the direction of indirect intention, and the estimated values do not cover the entire directional space. This results in poor performance of the FVAC method especially on circular paths. The Const. Rc method assumes indirect intention as a constant curvature curve on a plane, while in direct teaching, such intention might encompass varying curvatures constantly. Therefore, in both offline simulations and online experiments, the accuracy of the Const. Rc method in predicting human indirect intentions notably lags behind our proposed approach. Moreover, Const. Rc method, setting initial damping value as an inversely proportional function of velocity, can lead to abrupt damping decreases when velocity surpassed a minimum threshold, potentially causing unstable control. Further discussion is warranted regarding the remarkable enhancements achieved by our method. The estimation of direct intention solely discerns human intentions for rapid or precise movements, which could reduce the effort required by the operator. However, due to its

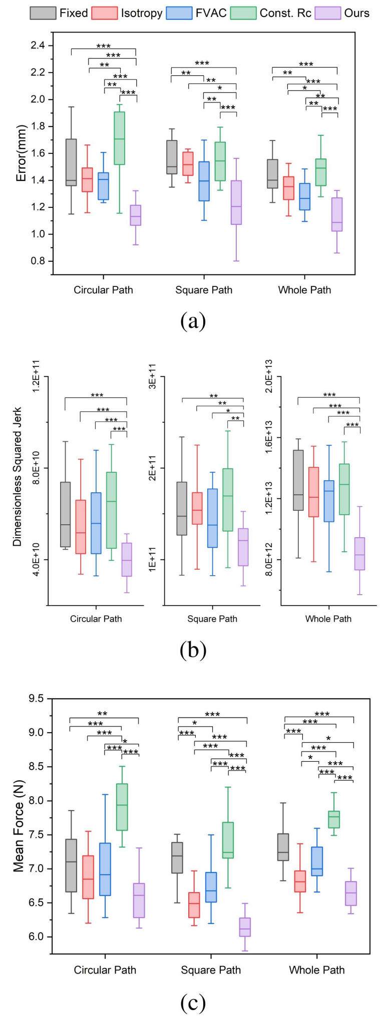

Fig. 8. Results of the online experiment task. The comparison is made according to the improvement of different metrics: (a) accuracy; (b) smoothness; (c) effort. Asterisks indicate level of statistical significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

isotropic nature, its effectiveness may be comparable to the isotropic contrast method, providing limited reduction in errors and smoothness enhancement. In contrast, the estimation of indirect intention assesses human trajectory and turning intentions, utilizing anisotropic modulation of the damping matrix to offer superior real-time haptic guidance in the intended direction. This approach can reduce human effort and better align with human expectations, thereby significantly improving trajectory accuracy and smoothness. In summary, the proposed controller integrates direct intention estimation with indirect intention estimation, achieving notable enhancements across the proposed metrics.

The proposed method still has some shortcomings. Our proposed method holds the potential for expansion into three-dimensional space. To simplify the experiment, following the approach of many prior studies in direct teaching, we validate our method in a two-dimensional plane, demonstrating its significant advantages over other methods. Furthermore, it has been shown that human motion in three-dimensional space is also consistent with the minimum-jerk model [30]. Also, when movements in three dimensions are considered, the power law applied in this letter is extended to speed-curvature-torsion power law [31]. This extended power law is demonstrated in surgical suturing when performed via RAMIS [32]. In [33], a new power law linking the speed to the geometry in human control on movement orientation is demonstrated by analyzing the movements commonly used in training of robotic surgeons. It can be expected that after the planar power laws have been verified in direct teaching, higher-dimensional power laws such as the power law in [31] and [33] can be further verified in future work to predict human indirect intent and provide better guidance to improve the performance of pHRI.

VI. Conclusion

In this study, a variable admittance controller is introduced based on both direct and indirect human intention. Minimum-jerk model is considered as the path-planning policy in human central nervous system and thus the velocity-curvature patterns are observed in planar movement. Based on these two theories, the estimation allows for the prediction of the next move and obtains the intended force. The variable admittance parameters are designed to guide the intended direction to compensate for the noise and errors. The effectiveness of the proposed scheme is evaluated by offline simulation and online experiments. The proposed scheme performs better with respect to the accuracy, smoothness and effort reduction of the trajectory when compared to fixed admittance control and state-of-the-art variable admittance control methods. The results show that the proposed variable admittance controller improves the intuitiveness and capability of pHRI.

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